

Winston Zhang
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Dear Hiring Manager,

I'm a Mechanical Engineering Penultimate year student with a strong foundation in mechanical engineering and hands-on experience in practical production and design, I believe I can contribute to your various multidisciplinary projects and further develop my professional expertise.

I believe my experience working on delivering an engineering product as part of a multidisciplinary team in Sydney Motorsports as well as my time as a CFD Intern for Dovetail Electric Aviation provides me with strong problem solving skills and hands-on experience in CAD, FEA, CFD as well as common manufacturing techniques will allow me to excel as part of the Northrop team. My key strengths include:

- **CFD and Thermal Analysis:** Experience in cooling duct as well as external aerodynamics simulation. I am also aware of and have conducted basic HVAC analysis for a restaurant environment using TRACE700 as well as fluent.
- **Mechanical Design and CAD:** I have extensive experience in SOLIDWORKS and Autodesk Inventor, with applications in aerodynamic structures construction and simulation, robotic systems and mechanical components.
- **Practical Engineering and Prototyping:** I have a strong foundation in composite manufacturing, mold fabrication and structural validation through numerical and physical testing.
- **Collaboration and Communication:** I have a proven ability to work in multidisciplinary teams, liaising with engineers, fabricators and non-engineering departments as the aerodynamic lead in Sydney Motorsports, which I believe will allow me to work well as part of Northrop's team.

I am excited about the opportunity to join your company and be part of an impactful project. Thank you for your consideration and I look forward to the opportunity to discuss my abilities and its alignment with your vision.

Sincerely,
Winston Zhang

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics)

Expected Graduation: 2025

2021 - Current

EXPERIENCE

SYDNEY MOTORSPORTS FORMULA STUDENT TEAM

Sydney, Australia

CAD | Aerodynamic Design | CFD Validation | Composite Manufacturing

Aerodynamics Engineer

Aerodynamics Section Lead

July 2022 - December 2023

December 2023 - February 2025

DOVETAIL ELECTRIC AVIATION

Sydney, Australia

CAD | CFD Validation | Reverse Engineering

Computational Fluid Dynamics Intern

September 2023 - January 2024

TECHNICAL SKILLS

Composite Manufacturing - Prepreg and Hand Layup

- **Hand Layup** - Fabricated the aerodynamic package for Sydney Motorsports including the nosecone, front wing, side-pods and rear wing.
- **Mold Manufacturing** - Manufactured the mold for the front wing and side-pods with the use of a laser cutter and a 3D printer.

Computer Aided Design - SOLIDWORKS and Autodesk Inventor

- **Parametric Design** - CAD for both Sydney Motorsports and Dovetail are fully parametric and dimensioned to be easily modified without breaking the larger assembly. Parts designed include:
 - **FSAE front wing** - General design as well as the rib and spar structure
 - **Nacelle and propeller** - Re-made the nacelle, propeller and cooling duct geometry from 3D scanned files for CFD simulations.
 - **Gearbox and general structure for a robot arm** - Designed a 6 axis robot arm from scratch.

Numerical Simulation - SOLIDWORKS and ANSYS

- **Semi-automated CFD Workflow** - Part of the development of the CFD workflow for Sydney Motorsports to enable streamlined sensitivity studies.
- **Topology Study** - Created optimized designs for the swannecks for Sydney Motorsport's wings as well as their internal rib structures.
- **Cooling Duct Optimization | Propeller Pitch Validation** - Optimized the cooling duct for Dovetail to minimize drag at design conditions. Validated the 3D scanned geometry of Dovetail's propeller geometry against the spec sheet.
- **Thermal Comfort** - Simulated a restaurant environment to assist with HVAC design to ensure the thermal comfort of the inhabitants.

Programming - MATLAB and Python

- **Matlab** - Blade Element Analysis, custom FEM and CFD code
- **Python** - Custom CFD code, Robot Operating System (ROS2)

TECHNICAL PROJECTS

- **Homelab Server** - Acts as a media server and custom cloud storage solution.
- **Robotic Arm** - 6 axis robotic arm with NEMA motors, belt drive and cycloidal gearbox.
- **Custom PC Peripherals** - Designed a 32 gram fingertip mouse and put together a lily58 split ortholinear keyboard and designed a case.